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Design of fiber-reinforced soft bending pneumatic artificial muscles for wearable tremor suppression devices

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Abstract

Soft robotics is a rapidly evolving field offering novel solutions in the development of wearable technologies. Soft pneumatic artificial muscles in particular, have seen widespread use in the development of human scale rehabilitative and assistive wearables. However, these soft actuators have not yet been adapted to address the complex dynamic regime of active (essential tremor) and resting (Parkinson's disease) hand tremor, the most common movement disorder affecting humans. Current solutions to address hand tremor involve expensive medication and surgical interventions, as well as wearable assistive devices that fall short of providing an effective compact design for the suppression of hand tremor. This study focuses on the design of a novel lightweight, compact, bending actuator that will be capable of actively suppressing hand tremor when adapted into an assistive wearable device. The proposed fiber-reinforced bending pneumatic artificial muscle (BPAM), including its design specifications, fabrication process, theoretical modeling, and experimental characterization, are detailed. The developed actuator was capable of producing sinusoidal trajectories with peak-to-peak amplitudes of 40° and a bandwidth of 8 Hz, the dynamic regime of pathological hand tremor. The ability of the fiber-reinforced BPAM to act within the dynamic regime of hand tremor demonstrates its potential to be further developed into a system capable of the active suppression of hand tremor.

Supplementary material for this article is available [online](#)

Keywords: pneumatic artificial muscles, soft bending actuators, soft wearable robots, fiber-reinforced polymers, tremor suppression

(Some figures may appear in colour only in the online journal)

1. Introduction

Through innovative electromechanical design and fabrication techniques, soft robotics is helping to push advancements in human-machine interactions forward. Specifically, the use of soft low rigidity materials, such as polydimethylsiloxane and silicone rubber [1–3] has allowed for creation of active and passive wearable mechanical devices that provide additional

functionality to the user while remaining compliant to the body [3–8]. Pneumatic artificial muscles (PAMs) [9–11] in particular have seen robust use in the design of wearable devices due to their light weight, compliance, and ability to produce linear, twisting, and bending motions when inflated [3, 4]. These traits have made PAMs a useful actuation system for the design of assistive and rehabilitative wearables to aid users who have limited joint mobility and/or muscular control.

However, PAMs have not yet been adapted to address the complex regime at which tremor, the most common movement disorder affecting the human body [12–14], occurs.

Tremor is characterized as an involuntary, rhythmic movement which manifests at the joints, inducing uncontrolled oscillatory deflections [14–17]. The most prevalent instances of this disorder may occur naturally, as in the case of physiological tremor, or as part of a medical syndrome, as in the cases of essential tremor (ET) and Parkinson's disease (PD). ET and PD are both characterized by large abnormal oscillations which occur during voluntary movement (i.e. action tremor) [14, 17–19] or during rest (resting tremor), respectively [14, 20, 21]. The symptomatic tremors associated with ET and PD are the most prominent in the joints of the hands and fingers (hand tremor). Hand tremor severely inhibits an individual's dexterity and ability to manipulate objects, which are essential to completing everyday tasks. Unfortunately, there currently are no cures to halt the progression of ET or PD. Although treatments exist to mitigate the effects of tremor, current options are very expensive; this has pushed researchers to search for alternative means of addressing hand tremors [22].

In recent years, there has been a rise in the development of inexpensive assistive devices designed to suppress hand tremor. Passive devices [23–25], such as the Read-Steady Glove [23] and GyroGlove [24], use the addition of passive weights to increase the inertia of the hand thereby reducing the effects of tremor. Active devices [26–30], such as the Emma Bracelet [26] and Liftware Steady™ Spoon [29], incorporate linear actuation systems or vibration to suppress tremor. These devices however, have a few drawbacks in their designs. The added inertia of passive systems limits maneuverability and do not sufficiently address tremors in the finger joints. Meanwhile, the active devices are typically task specific and/or bulky, conspicuous, and rigid, inhibiting mobility and drawing unneeded attention to users.

This study focuses on the development of a novel, lightweight, compliant, and inconspicuous fiber-reinforced bending pneumatic artificial muscle (BPAM) that can act as an active actuation system when integrated into a future assistive glove device. Unlike current rehabilitative gloves [6, 31–34] and tremor suppression devices [23–30], the developed BPAM was designed to actively suppress tremor by counteracting the dynamic motions and trajectories, including angular deflections/amplitudes and oscillating frequencies, associated with ET and PD tremor. The fiber-reinforced BPAM design has not been previously fabricated, nor has it been characterized experimentally or through theoretical modeling, which are both done in this study. In addition, the potential for the BPAM to produce trajectories capable of counteracting hand tremor were demonstrated through trajectory tracking tests of simple sinusoidal inputs conducted on a healthy subject's index finger. This demonstration was conducted to show the potential of the actuator to be used in a future device for the suppression of hand tremor.

In order to address the shortcomings of traditional cylindrical PAMs, a two-dimensional (2-D) flat PAM is proposed. This study focuses on the design and theoretical modeling of

the BPAMs. Furthermore, the proposed flat muscle design was realized through a simple layered fabrication method, and the theoretical model was experimentally verified. The remainder of the study is organized as follows: section 2 details the design and fabrication of the proposed pneumatic bending actuator; section 3 discusses development of a theoretical model for the actuator; section 4 details the experimental setup and provides analysis on the experimental results; section 5 finally concludes the work with discussion of future work.

2. Design

2.1. Design specifications

A novel fiber-reinforced BPAM, shown in figure 1, was designed to address the dynamic characteristics of hand tremor while remaining streamlined and compact when compared to the dimensions of the human hand. In order to achieve this, the following specifications needed to be met.

2.1.1. Dimensions. Similar to active soft glove devices developed for rehabilitative purposes [6, 31–33], the actuator developed in this study was designed to fit along the back of the human fingers. With the authors' fingers used as the basis for prototyping, the actuator had to accommodate (i) a maximum finger length of 90 mm in order to ensure actuation along the entire length of the fingers and their joints; (ii) a width of less than 16.5 mm, to ensure that the BPAM could rest on the top of the finger without interfering with mobility; and (iii) a thickness of less than 5.5 mm, or half the minimum measured finger thickness (11 mm) within the group, to minimize the bulkiness of the system in relation to the overall finger dimensions. The proposed thickness specification is much smaller than that of previous soft actuators developed for wearables which are in the range of 1–2 cm [6, 30–32].

2.1.2. Actuation. In order for the developed BPAM to act as an active suppressor of hand tremor, the actuator needed to operate within a dynamic range equivalent to or exceeding that of pathological tremor. ET and PD, are characterized by easily visible oscillations (amplitudes of 0° – 20°) in the range of 4–8 Hz [17, 20, 35]. PTs are characterized by 8–12 Hz oscillations, typically of small amplitudes (0° – 2.5°), making them invisible to the naked eye [16, 17, 35]. Given the dynamic ranges of tremor, the developed actuator would have to be capable of 4–12 Hz oscillations of 0° – 20° amplitudes.

The bending tip force characteristics are estimated from the range of forces generated by human fingers during active flexion and extension. The digitorum profundus (DP) and superficialis tendons (SFT) are major muscle groups that flex the fingers. The DP and SFT are able to generate forces of 1.3–4 N and 1.3–8.5 N, respectively [36]. This translates to a bending tip force of 1–4 N when conducting grasping motions while eating [37]. Because the developed actuator is aimed at counteracting tremor, and not providing bending assistance, we assumed that a maximum bending tip force of 1–2 N

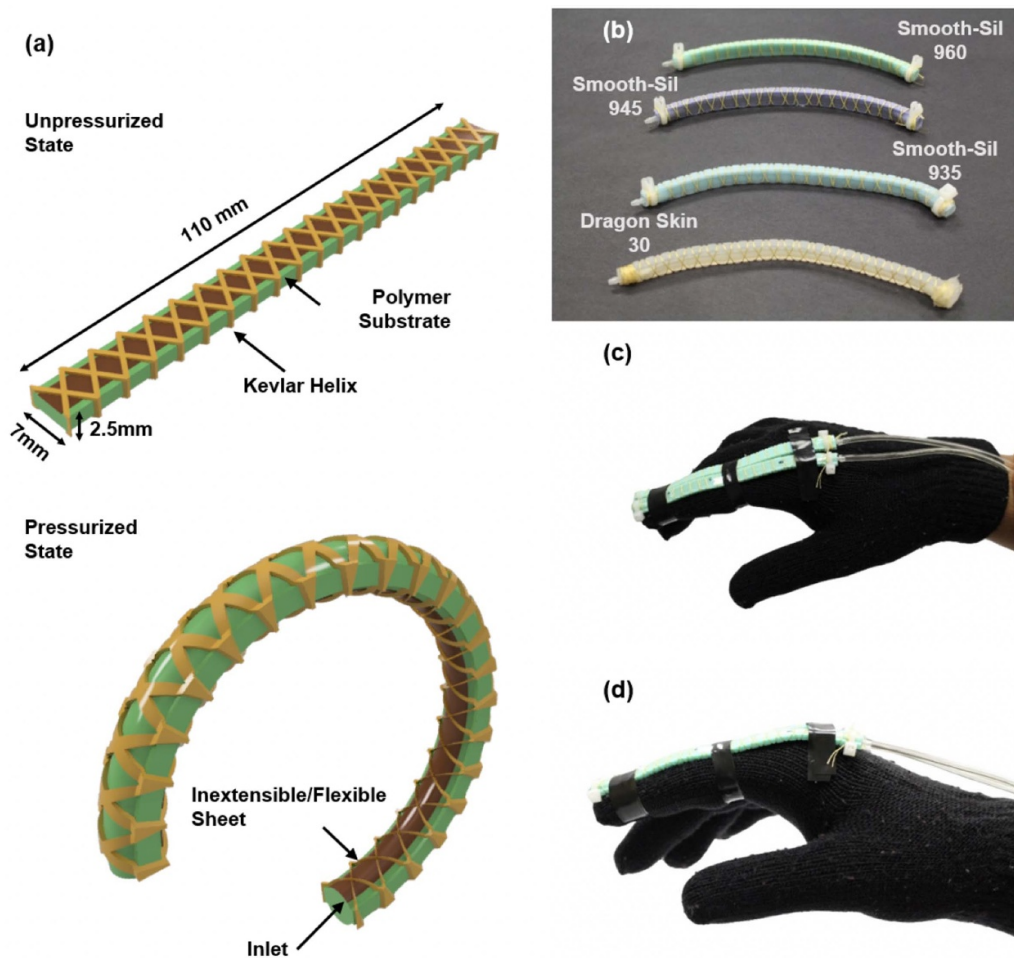


Figure 1. Flat pneumatic muscles and their integration to wearable device. (a) Solid model of completed BPAM with labels designating key components, (b) Fabricated fiber-reinforced BPAMs composed of Smooth-Sil 960 (green), Smooth-Sil 945 (purple), Smooth-Sil 935 (blue), and Dragon Skin 30 (transparent). (c) Top view of potential actuator-glove configuration on human hand, (d) Side view of potential actuator-glove configuration on human hand.

would be sufficient. This would allow the actuator to generate 25%–50% of the maximum observed tip forces.

2.2. Actuator design

As seen in figure 1, the fiber-reinforced BPAMs were composed of a thin elastomeric core containing a collapsed ‘zero-volume’ air chamber (ZAC). The core section was fabricated from various silicone rubbers from Smooth On Inc. including Dragon Skin 30 ($E_{100} = 0.59$ MPa), Smooth-Sil 935 ($E_{100} = 1.17$ MPa), Smooth-Sil 945 ($E_{100} = 1.79$ MPa), and Smooth-Sil 960 ($E_{100} = 1.93$ MPa). Here E_{100} is the manufacturer reported tensile modulus of the silicone rubber materials at 100% strain. The thin rectangular form factor of the core was chosen to allow the BPAM to remain compact in its rest state and during operation. A single laser cut plastic sheet (PEI Ultem, McMaster Carr) was secured to the core section by winding Kevlar fibers at alpha and beta helix angles of 75° (Kevlar 69 thread, McMaster Carr) around the outer surfaces of the actuator. A high volume-fraction of Kevlar fibers was used in the completed actuator

in order to geometrically constrain the deformed form factor of the actuator when inflated with pressurized air [1, 2, 7, 35]. The fibers served to restrict the deformed thickness of the actuator, ensuring the system would remain compact during operation. The plastic sheet provided an additional geometric constraint to the system, which was integral in forcing the actuator to bend when inflated. By attaching the sheet to a single side of the BPAM, the deformed length along that side was constrained to its original length. The opposite side of the actuator however, was free to elongate along its length. This constraint on the first side coupled with elongation of the opposite side caused the actuator to bend with increasing pressure input.

The final dimensions of the BPAM were $110 \text{ mm} \times 7 \text{ mm} \times 2.5 \text{ mm}$ thick, with ZAC dimensions of $110 \text{ mm} \times 4 \text{ mm} \times 0.1 \text{ mm}$ thick and weight about 9.5 grams. When inflated, the overall actuator thickness increased to 5–8 mm depending on the material used to create the elastomeric core. The developed BPAM at 2.5 mm thick and 8.5 grams is much thinner and of equal or lesser weight than actuators found in previous soft gloves [31–33]. This form factor provides a minimal increase

to the overall diameter of the fingers, and should not cause excessive fatigue due to its light weight. Additional details regarding the development and design of BPAM typed actuators can be found in the thesis by Wirekoh [38, 39].

2.3. Fabrication

Initially, the thin elastomeric core section was fabricated using 3D printed molds (VeroBlack, Stratasys Ltd.), liquid-cure silicone rubber, and a water-soluble mask (Water Soluble Stabilizer, Paper Solvy). Unlike collapsed chamber actuators found in the literature [5] which are fabricated by spraying an anti-bonding agent such as mold release, the use of the water-soluble mask allows the BPAMs to be formed from a single cured substrate of liquid elastomer rather than bonding two separate sheets together [38, 39]. This improves repeatability in the fabrication process and robustness by preventing the onset of unexpected delamination. The removal of the water-soluble mask by flushing water through the core several times, then allowed for the creation of the collapsed air chamber, e.g. ZAC [5]. Following this step, an inlet nozzle and stopper were added to the two ends of the BPAM. Next, a single laser cut plastic sheet was secured to the core section by winding Kevlar fibers at alpha and beta helix angles of 75° along the outer surfaces. The Kevlar fibers and plastic sheet provided geometric constraints that caused the actuator to bend when inflated. The actuator was then completed by utilizing zip-ties to ensure that the tube fittings, plastic sheet, and fibers were firmly secured in place. Figure 2 provides additional information regarding fabrication of the proposed BPAM.

3. Theoretical modeling

In order to approximate the behavior of the proposed fiber-reinforced BPAM, a theoretical model was developed. As seen in figure 3, the BPAM was modeled as two flat rectangular composite plates simply supported along their edges and free to bend with dimensions $2a \times 2b \times 4h$. The actuator could be modeled as two composite plates because the air volume is created by layering two elastomeric substrates. The model incorporated a modified von Karman's formulation for large deformations [38, 39], utilizing superimposed solutions for simply supported plates and plates undergoing pure bending. Geometric constraints similar to those imposed in [1, 2, 7] for the design of fiber-reinforced PAMs using helical patterns were also incorporated. Subsequently, the energy method was used to find an analytic solution to the modified von Karman's formulation for the large deformation of plates [38–41]. The formulation begins with the elastic model developed in [38] in order to approximate the mechanical performance of the BPAM. Assuming an initial z -axis deflection

$$w(x, y) = w_0 \cos \frac{\pi x}{2a} \cos \frac{\pi y}{2b}, \quad (1)$$

as postulated by Timoshenko [40, 41], and solving the modified von Karman's formulations for a composite plate

$$\begin{aligned} & \frac{1}{E_y} \frac{\delta^4 \phi}{\delta x^4} + \frac{1}{E_x} \frac{\delta^4 \phi}{\delta y^4} + \left(\frac{1}{G_{xy}} - \frac{\nu_{xy}}{E_x} - \frac{\nu_{yx}}{E_y} \right) \frac{\delta^4 \phi}{\delta x^4} \\ & = \left[\left(\frac{\delta^2 w}{\delta x \delta y} \right)^2 - \frac{\delta^2 w}{\delta x^2} \frac{\delta^2 w}{\delta y^2} \right], \end{aligned} \quad (2)$$

yielded the following form for the x - and y -axis displacements:

$$u(x, y) = \frac{u_0 x}{2a} + u_1 \sin \left(\frac{\pi x}{a} \right) + u_2 \cos \left(\frac{\pi y}{b} \right) \sin \left(\frac{\pi x}{a} \right), \quad (3a)$$

$$v(x, y) = \frac{v_0 y}{2b} + v_1 \sin \left(\frac{\pi y}{b} \right) + v_2 \cos \left(\frac{\pi x}{a} \right) \sin \left(\frac{\pi y}{b} \right). \quad (3b)$$

Here u , v , and w are displacements in x -, y -, and z -axis directions, respectively. The subscripted displacement components (e.g. u_0 , v_0 , v_1 , w_0 , etc.) are unknown constant values that will be solved for through the minimization of the systems potential energy, which will be expanded on later in this section. Equations (1) and (3) provide solutions in the case of pure deformation of the actuator in the plane (i.e. the plates have no z -axis deflection along their edges). In order to model the bending deformation, a bending deflection was devised using a similar form to that postulated by Timoshenko. In this case, where pure bending occurred, the z -axis deflection was assumed to take on the form

$$w(x, y) = w_1 \cos \frac{\pi x}{2a}. \quad (4)$$

The assumed deflection has no dependence on y , but has z -axis deflection at the maximum x -axis dimensions of the plate ($x = a$). Solving equations (2) using (4) yielded the following x - and y -axis deflections:

$$u(x, y) = \frac{u_0 x}{2a} + u_1 \sin \left(\frac{\pi x}{a} \right), \quad (5a)$$

$$v(x, y) = \frac{v_0 y}{2b}. \quad (5b)$$

With knowledge that the actuators both bend and deform during operation, a superimposed formulation was used to solve for the deformation of the system. The total effects of the deflections of the two plates were specified by combining the displacements of both pure bending and pure deformation such that

$$w(x, y) = w_{\text{deform}} + w_{\text{bend}}, \quad (6a)$$

$$u(x, y) = u_{\text{deform}} + u_{\text{bend}}, \quad (6b)$$

$$v(x, y) = v_{\text{deform}} + v_{\text{bend}}, \quad (6c)$$

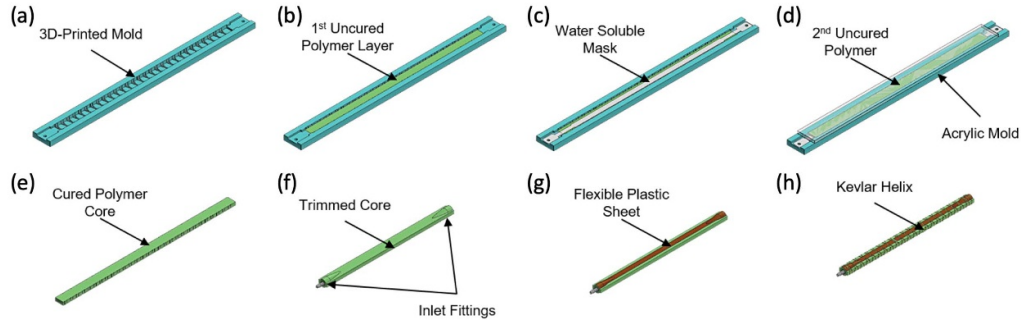


Figure 2. Fabrication process of BPAM using water-soluble mask: (a) prepare bottom core mold, (b) pour liquid elastomer for bottom half of core, (c) place water-soluble mask, (d) fill top half of mold and place acrylic cap, (e) after curing, remove core from mold, and open air volume using water to remove mask, (f) cut actuator to length, then insert tube fittings, (g) align flexible/inelastic plastic sheet, and (h) secure the plastic sheet by winding the fibers, followed by cutting the plastic sheet to allow extension at the joints.

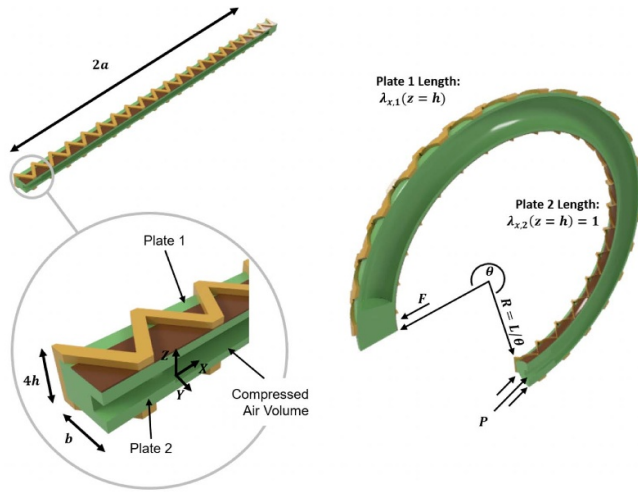


Figure 3. Schematic of BPAM depicting coordinate system used to develop the theoretical model.

$$\lambda_x = \varepsilon_{x,deform} + \varepsilon_{x,bend} + 1, \quad (6d)$$

$$\lambda_y = \varepsilon_{y,deform} + \varepsilon_{y,bend} + 1, \quad (6e)$$

$$\varepsilon_{xy} = \varepsilon_{xy,deform} + \varepsilon_{xy,bend}. \quad (6f)$$

The formulations in equation (6) could then be further specified for each of the plates as,

$$w_i(x, y) = w_{i0} \cos \frac{\pi x}{2a} \cos \frac{\pi y}{2b} + w_{b,i1} \cos \frac{\pi x}{2a}, \quad (7a)$$

$$u_i(x, y) = \left[\frac{u_{i0}x}{2a} + u_{i1} \sin \left(\frac{\pi x}{a} \right) + u_{i2} \cos \left(\frac{\pi y}{b} \right) \sin \left(\frac{\pi x}{a} \right) \right] + \left[\frac{u_{b,i0}x}{2a} + u_{b,i1} \sin \left(\frac{\pi x}{a} \right) \right], \quad (7b)$$

$$v_i(x, y) = \left[\frac{v_{i0}y}{2b} + v_{i1} \sin \left(\frac{\pi y}{b} \right) + v_{i2} \cos \left(\frac{\pi x}{a} \right) \sin \left(\frac{\pi y}{b} \right) \right] + \left[\frac{v_{b,i0}y}{2b} \right], \quad (7c)$$

where i is the index equal to 1 (top plate) or 2 (bottom plate) as seen in figure 3. The subscripted displacement components (e.g. u_{i0} , v_{i0} , v_{i1} , w_{i0} , etc.) are unknown constant values that will be solved for through the minimization of the system's potential energy. A nonlinear hyperelastic strain energy density, W , with volumetric stretch ratio J , was then incorporated from [38]

$$W_i = \frac{E}{6} \left(\lambda_{xi}^2 + \lambda_{yi}^2 + \lambda_{zi}^2 + \frac{\varepsilon_{xyi}^2}{2} - 3 \right), \quad (8a)$$

$$J_i = \lambda_{xi} \lambda_{yi} \lambda_{zi} - \frac{\varepsilon_{xyi}^2}{4} \lambda_{zi} \quad (8b)$$

for an incompressible Neo-Hookean solid. Here, the tensile modulus E , was the effective hyperelastic modulus of the composite actuator determined through experimental measurements. The normal stretches λ_x in the x -axis direction, λ_y in the y -axis direction, and shear strain ε_{xy} were then defined as,

$$\lambda_{xi} = \left[\frac{\delta u_i}{\delta x} + \frac{1}{2} \left(\frac{\delta w_i}{\delta x} \right)^2 - z \frac{\delta^2 w_i}{\delta x^2} \right] + \left[\frac{\delta u_{b,i}}{\delta x} + \frac{1}{2} \left(\frac{\delta w_{b,i}}{\delta x} \right)^2 - (z+h) \frac{\delta^2 w_{bi}}{\delta x^2} \right] + 1, \quad (9a)$$

$$\lambda_{yi} = \left[\frac{\delta v_i}{\delta y} + \frac{1}{2} \left(\frac{\delta w_i}{\delta y} \right)^2 - z \frac{\delta^2 w_i}{\delta y^2} \right] + \left[\frac{\delta v_{b,i}}{\delta y} + \frac{1}{2} \left(\frac{\delta w_{b,i}}{\delta y} \right)^2 - (z+h) \frac{\delta^2 w_{bi}}{\delta y^2} \right] + 1, \quad (9b)$$

$$\varepsilon_{xyi} = \left[\frac{\delta u_i}{\delta y} + \frac{\delta v_i}{\delta y} + \frac{\delta w_i}{\delta x} \frac{\delta w_i}{\delta y} - 2z \frac{\delta^2 w_i}{\delta x \delta y} \right] - \left[\frac{\delta u_{b,i}}{\delta y} + \frac{\delta v_{b,i}}{\delta y} + \frac{\delta w_{b,i}}{\delta x} \frac{\delta w_{b,i}}{\delta y} - 2(z+h) \frac{\delta^2 w_{b,i}}{\delta x \delta y} \right], \quad (9c)$$

where w , u , and v are displacement formulations due to pure deformation and w_b , u_b , and v_b are the displacement formulations for pure bending. The bending strains through the thickness of the plates were defined with this form to account for the fact that the two plates must maintain contact at their edges during bending. This edge constraint manifests in the following boundary constraints:

$$w_{21} = -w_{11} \text{ (due to coordinate system of plates)}, \quad (10a)$$

$$u_{20} = u_{10}, \quad (10b)$$

$$v_{20} = v_{10}, \quad (10c)$$

$$u_{b,20} = u_{b,10}, \quad (10d)$$

$$v_{b,20} = v_{b,10}. \quad (10e)$$

In addition, two geometric constraints were defined. First, the actuator had to maintain a constant length at $z_2 = h$ due to the location and properties of the plastic sheet. This constraint was formulated using an average stretch

$$1 = \frac{1}{2a \cdot 2b} \int_{-b}^b \int_{-a}^a \lambda_{x2}(z = -h) dx dy. \quad (10f)$$

Secondly, a geometric constraint associated with the fiber windings was incorporated as in [31–33] at $z = h$ for both plates. This constraint was given as

$$\frac{w + \lambda_{yi} 2b}{w + 2b} = \frac{\text{sgn}(\beta) \sqrt{1 - \cos(\beta)^2 \lambda_{xi}^2} \cos(\alpha) - \text{sgn}(\alpha) \sqrt{1 - \cos(\alpha)^2 \lambda_{xi}^2} \cos(\beta)}{\sin(\alpha - \beta)} \quad (10g)$$

where w is the distance the fibers wrap around the undeformed sections of the actuator per loop. The incompressibility of the plates resulted in a volumetric constraint caused by the deformation component of the plates' deflection which was defined as

$$v_{i0} = -\frac{2u_{i0}b}{(2a + u_{i0})}. \quad (10h)$$

Finally, a constraint was placed on the maximum x - and y -axis deflection such that the deformed actuator configuration was

realistic. This constraint was defined such that the ends of the actuators would come into contact at a bending angle of 360° and that bending would have negligible effects on strain in the y -axis direction as,

$$u_{b,i0} = -\frac{\theta}{\pi} a. \quad (10i)$$

$$v_{b,i0} = 0. \quad (10j)$$

Utilizing the constraints in equation (10), the potential energy of the system was defined as

$$\Pi = \iiint W_1 dV_1 + \iiint W_2 dV_2 - W_{P1} - W_{P2} - W_m, \quad (11)$$

where $\iiint W_i dV_i$ is the strain energy of each plate, $W_{P_i} = \iint^P [w_{i0} \cos \frac{\pi x}{2a} \cos \frac{\pi y}{2b} + w_{b,i1} \cos \frac{\pi x}{2a}] dx dy$ is the work done by pressure to deform the plates, and $W_m = M\theta$ is the work done by the acting moment, with the maximum moment being defined as $M_{max} = 2a * F$. Π was then minimized with respect to the unknowns u_{11} , u_{12} , v_{11} , v_{12} , u_{21} , u_{22} , v_{21} , v_{22} , $w_{b,11}$ and θ (i.e. $\frac{\delta \Pi}{\delta \theta} = 0$) to establish a relationship between the required input pressure, P , and the desired bending angle θ . The theoretical model was then compared to experimental results.

4. Experiments

4.1. Experimental setup

The fiber-reinforced BPAMs were subjected to various experiments in order to characterize their mechanical performance. The experimental apparatus (figure 4) employed was composed of four proportional solenoid valves (MDPRO5VCF8S 11.5 VDC, Parker), two industrial pressure sensors (#1032663, BangGood) capable of reading up to 150 psi (approximately 1 MPa), an Arduino Uno microcontroller (Arduino), a force sensor (FSAGPDXX001RCAB5, Honeywell) capable of reading up to 10 N, a custom microfluidic soft sensor designed to measure bending angle of the actuators, and custom electronics designed to effectively power the system.

Several tests were conducted to establish how the BPAMs behaved during operation. Initially, the BPAMs' maximum bending angle at various input pressures was examined. During this procedure, the BPAMs were secured to a flat surface, then pressurized with compressed air, as seen in figure 4(a). The actuators were inflated up to a maximum angular deflection of 270° (physical limitation of experimental setup) where possible. The angular deformation was captured by the soft angle sensor and recorded using the microcontroller.

In the next procedure, the dynamic properties of the BPAMs were examined. The actuators were again fixed to a flat surface at one end. The actuators were then cycled from 0 to their maximum experienced pressure during the angle experiments. This experiment was used to dynamically characterize the activation time, deactivation time, and bandwidth of the actuators. Cycles occurred at frequencies of 0.25–12 Hz, which

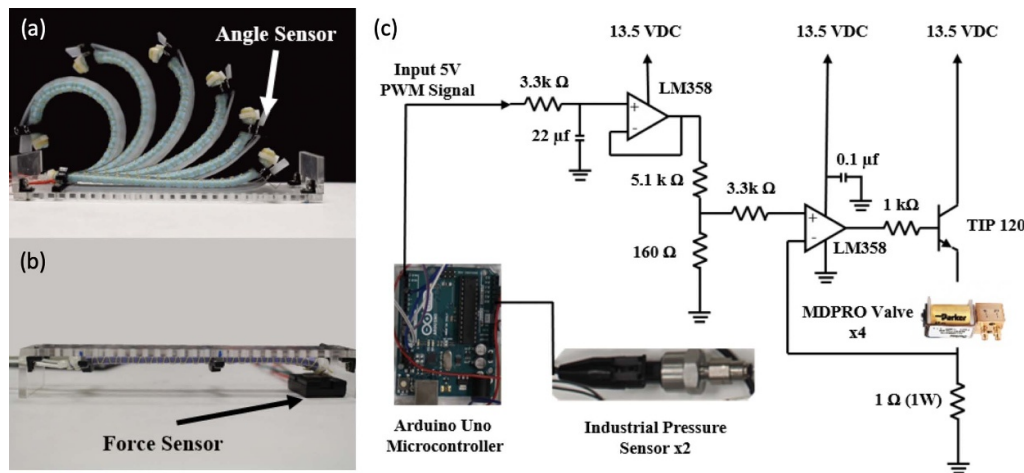


Figure 4. Diagram of experimental apparatus used to characterize the prototype fiber-reinforced BPAMs. (a) Overlay of the bending deformation of an BPAM during angle characterization experiments. (b) BPAM positioning during force control experimentation. (c) Schematic of electronic components utilized throughout experimentation.

are equivalent to or below the dynamic range of hand tremor [16, 17, 20, 33].

Subsequently, experiments were conducted to measure the maximum tip force generated by the BPAMs at input pressure. During this procedure, the actuators were fixed against a flat surface at three points as shown figure 4(b). The actuators were then pressurized using the same pressure ranges used for the bending experiments. Weights were used to ensure contact was maintained between the tip of the actuators and the force sensor during experimentation. ‘No load’ characterization experiments were conducted not only to provide a baseline for the maximum expected angular deflections and bending forces at given input pressures, but also to reflect experimentation conducted on the characterization of similar actuators in the literature [1, 32, 42, 43].

Following the conclusion of primary experimentation to characterize the developed actuators, a demonstration of the BPAM’s potential to actively produce trajectories within the dynamic regime of hand tremor was conducted. During the demonstration, the BPAMs were secured, in antagonistic pairs, to the resting index finger of a healthy person. The actuators were then tasked with generating sinusoidal trajectories, within the dynamic regime of hand tremor (e.g. produce a sinusoidal trajectory with a peak-to-peak amplitude of 0–40° at a frequency of 1–8 Hz) [16, 17, 20, 33]. By attaching the BPAMs to a human finger, it became possible to observe if the actuators could produce required amplitude and frequency oscillations despite the added resistance from live tissues. Ultimately, the goal of this demonstration was to observe if the proposed actuators could be feasibly work in the required regime needed to actively suppress pathological hand tremor when integrated into a future wearable device.

4.2. Results and discussion

The four fiber-reinforced BPAMs developed in this study, which were composed of Dragon Skin 30, Smooth-Sil 935, Smooth-Sil 945, and Smooth-Sil 960, were experimentally

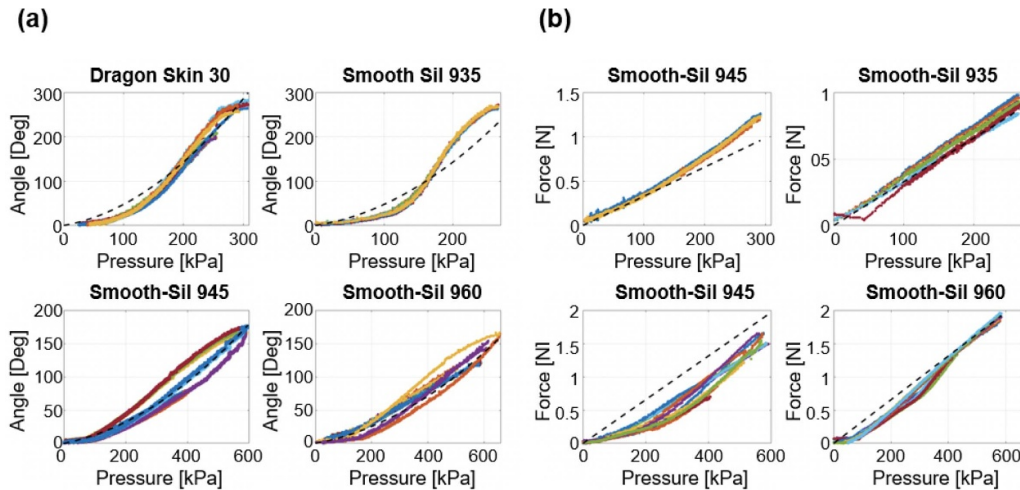
characterized to determine their maximum bending tip force and bending angle when pressurized. The effective hyper-elastic moduli for the fiber-reinforced BPAMs created from the four silicone rubber materials can be found in table 1. In addition, the maximum bending angles and generated forces can be found in figures 5(a) and (b) respectively, with additional results found in table 1.

The developed actuators were able to reach maximum bending angles far larger than the required 40° peak-to-peak range associated with ET and PD, figure 5 and table 1. The actuators composed of Smooth-Sil 945 (SS-945) and Smooth-Sil 960 (SS-960) reached bending angles of $176.7 \pm 2.4^\circ$ and $164.6 \pm 4.1^\circ$, respectively, while those composed of Dragon Skin 30 (DS-30) and Smooth-Sil 935 (SS-935) reached bending angles of $282.9 \pm 1^\circ$ and $272.3 \pm 1.5^\circ$ respectively. The former two actuators could further reach larger bending angles, if not for the constraint of the experimental setup which limited the maximum deflection of the actuators. In addition, the BPAM’s fabricated from DS-30, SS-935, SS-94, and SS-960, generated maximum overall forces of 1.26 ± 0.05 N, 0.98 ± 0.12 N, 1.65 ± 0.22 N, and 1.96 ± 0.06 N, respectively. During experimentation the actuators were found to fail above 345 kPa for DS-30 and SS-935 and above 655 kPa for SS-945 and SS-960. Typically, these failures were due to the wound fibers shearing through the actuators at the corners, or were pinhole failures occurring around the inlet fittings due to the material being stretched to accommodate the cylindrical tube fittings.

The mechanical performance for each of the prototype actuators was estimated using the developed model. The hyper-elastic modulus for each actuator was passed as an input to equation (11), in order to predict their bending deformation and generated tip force. These estimated values were then compared to experimental results (as seen in figure 5) and the RSME errors were calculated. The developed theoretical model compares favorably to the mechanical performance of the fiber-reinforced BPAMs measured experimentally, having an average root-mean-squared error (RMSE) of $15.8 \pm 9^\circ$ and

Table 1. Performance and Theoretical Comparison of BPAMs.

Material	Effective Modulus (MPa)	Max. Angle (°)	Max. Force (N)	Theoretical Angle RMSE (°)	Theoretical Force RMSE (N)
Dragon Skin 30	3.558	282.9 ± 1	1.26 ± 0.05	20.2	0.07
Smooth-Sil 935	3.310	272.3 ± 1.5	0.98 ± 0.12	26.15	0.06
Smooth-Sil 945	9.308	176.7 ± 2.4	1.65 ± 0.22	10.36	0.32
Smooth-Sil 960	10.963	164.6 ± 4.1	1.96 ± 0.06	6.43	0.18

**Figure 5.** Mechanical performance of the four fiber-reinforced BPAMs fabricated. (a) Maximum experimental bending angle measured during experimentation. (b) Maximum generated bending tip force measured at zero angle deflection. (Model prediction is shown with a dotted line in each plot).

0.16 ± 0.13 N for its approximation of the bending angle and tip force of the developed actuators, respectively. When compared to the maximum overall performance of the actuators, this amounts to an RMSE ratio of $6.7 \pm 2\%$ and $9.9 \pm 7\%$ for the force and bending angle, respectively. This error can partially be attributed to the SS-935 material, which behaves particularly differently from the other silicone rubbers utilized in this study. In addition, poor sealing of the actuators led to air leakage at higher pressure inputs. However, due to the model's ability to approximate the trajectory and performance of the BPAM's, researchers should be able to utilize it to predict the performance for future designs.

Examples of the dynamic performance of the actuators are shown in figure 6. The actuators had activation times of 0.159 ± 0.005 s and deactivation times of 0.278 ± 0.023 s. These times were evaluated as the time required to reach the maximum angular deformation, which was about 180° for the softer materials (SS-945 and SS-960) and 270° for the stiffer materials (DS-30 and SS-935). This resulted in an average activation and deactivation speed of $1415^\circ/\text{s}$ (40° displacement at 35.4 Hz) and $820^\circ/\text{s}$ (40° at 20.2 Hz) respectively, which was sufficient to address the range and the amplitude of hand tremors. In this study, the bandwidth was defined as the cutoff frequency at which the actuators were unable to produce a 40° peak-to-peak amplitude (the maximum joint deflection associated with PD and ET), when given a sinusoidal input signal with an amplitude corresponding to the

actuators' maximum angular deformation (180° for SS-945 and SS-960 or 270° for DS-30 and SS-935). With this definition for bandwidth, the cutoff frequency was around 8 Hz, the maximum frequency typically seen in ET and PD tremor. Between the ranges of 4–8 Hz, the actuators displayed a maximum phase lag of $13 \pm 0.7^\circ$. A phase lag within this regime should not adversely affect the implementation of a control scheme that actively produces antagonistic trajectories when tremor is detected. However, future research must consider the impact of phase lag and bandwidth in choosing optimal valves and active control schemes. Developing a control algorithm to provide bi-directional force (flexion and extension) can add additional rigidity to the fingers when tremor is detected. In this manner, the actuators can act as a lightweight variable stiffness stabilize to suppress tremor.

Examples of trajectories measured during the demonstrations with a healthy human subject can be found in figure 7. When attached to a human finger, the BPAMs were able to generate oscillatory behavior with maximum peak-to-peak amplitudes between 7.5° and 15° at frequencies of 1–8 Hz. Although the BPAMs could produce large angular displacements at frequencies between 4 and 8 Hz when unloaded, the added stiffness of the finger reduced the overall displacement of the actuators. However, this range of motion is still fairly significant, displaying the potential for the developed actuators to be integrated into a future wearable device.

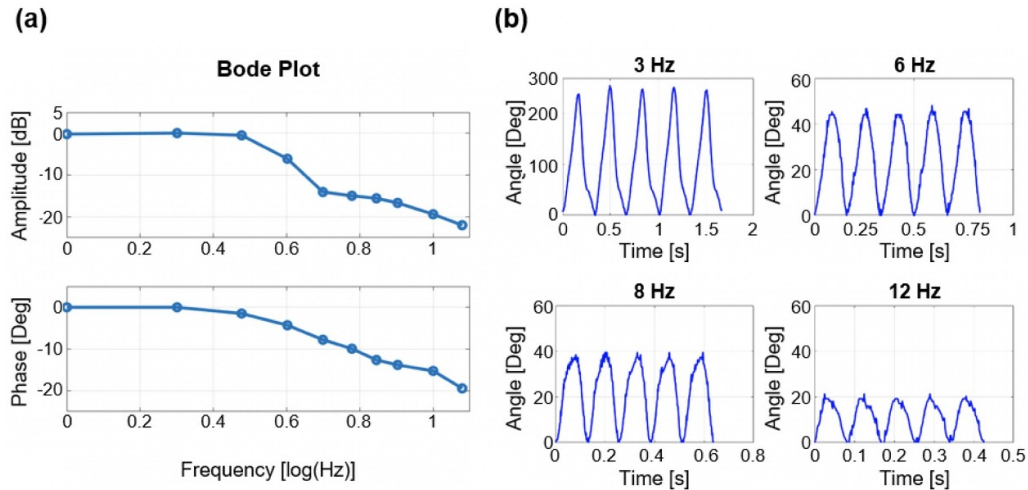


Figure 6. Dynamic performance of prototype BPAMs. (a) Bode plot of actuators at frequencies up to 12 Hz. (b) Examples of the dynamic response of the actuators at various frequencies. Max amplitude of 270° for DS-30 and SS-935 and max amplitude of 180° for SS-945 and SS-960.

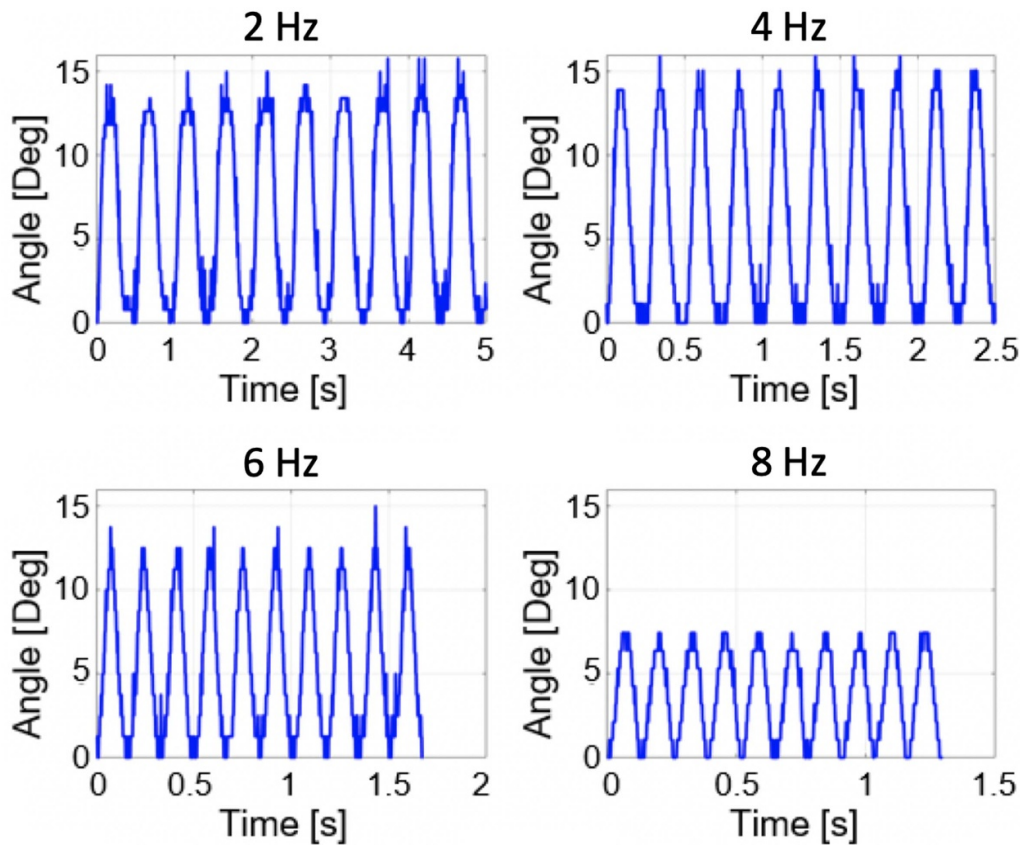


Figure 7. Oscillatory trajectories produced by antagonistic BPAM pair when connected to a human index finger. (The actual motions can be seen in the supplementary video(available online at <https://stacks.iop.org/SMS/30/015013/mmedia>)).

5. Conclusion and future work

A fiber-reinforced bending flat PAM was designed with the specifications required to actively suppress hand tremor associated with ET and PD. The mechanical performance of the prototyped BPAM was experimentally characterized through both static and dynamic tests with actuators manufactured

from a variety of silicone rubbers. In addition, a theoretical model was developed to approximate the bending angle and force generation of the actuator at input pressure and showed close approximation when compared to experimental characterization of the actuator. Ultimately, the actuators were capable of producing 40° peak-to-peak amplitudes in the required ET and PD frequency range of 4–8 Hz. In addition,

demonstrations were conducted to show the potential of the sleek lightweight actuator to produce trajectories capable of addressing hand tremor when to a human finger. However, this is only a first step, and additional improvements should be made in order to address hand tremors.

When compared to other pneumatic bending actuators designed for assistive devices [1, 31, 34], the BPAM designed in this study stands out for its slim design, which is only 2.5 mm thick when not inflated while most pneumatic designs for assistive devices have initial diameters of 1–2 cm. However, we plan to further miniaturize our design similar to thin pneumatic muscles, such as Peano muscles [44].

Further considerations for future work include the improvement of actuator performance in two key areas. First, the actuators should be fabricated with elliptical edges to minimize the occurrence of failure due to the fibers shearing through the material along the corners of the actuators. Second, the bending force generated by the actuators should be improved through a redesign and/or material selection in order to allow greater displacement of the finger at high frequencies. One potential redesign is by fabricating the antagonistic actuator pair as a stacked singular actuator instead. In this way, angular bending and extension can be combined into a single actuator that fits over the back of the finger with the capability of producing larger bending moments due to the increased width of the air volume. Furthermore, fatigue life testing for the improved design should be conducted, as actuators will be continually cycled during their operational lifetime. In addition, custom tube fittings should be designed to better fit the form factor of the BPAMs. Subsequently, a full glove system should be designed, to better house the actuators and provide improved transmission of the actuator force to the fingers during operation. The future wearable will then need to be tested with ET and PD symptomatic patients.

Another area of future work will be controlling the natural frequency of the actuator using variable stiffness materials or mechanisms, such as jamming structures [45, 46], conductive thermoplastic elastomer [47], or low melting point alloys [48]. The increase of the actuator's stiffness will increase the natural frequency, which will help customizing the device to a wide range of potential users with hand tremor.

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References

- [1] Polygerinos P, Wang Z, Kevin G C, Wood R J and Walsh C J 2015 Soft robotic glove for combined assistance and at-home rehabilitation *Robot Auton. Syst.* **73** 135–43
- [2] Bishop-Moser J and Kota S 2015 Design and modeling of generalized fiber-reinforced pneumatic soft actuators *IEEE Trans. Robot* **31** 536–45
- [3] Rich S I, Wood R J and Majidi C 2018 Untethered soft robotics *Nat. Electron.* **1** 102–12
- [4] Zhang J, Sheng J, O'Neill C T, Walsh C J, Wood R J, Ryu J-H, Desai J P and Yip M C 2019 Robotic artificial muscles: current progress and future perspectives *IEEE Trans. Robot* **35** 761–81
- [5] Park Y-L, Santos J, Galloway K G, Goldfield E C and Wood R J 2014 A soft wearable robotic device for active knee motions using flat pneumatic artificial muscles *Proc. IEEE Conf. Robot Autom. (ICRA) Hong Kong, China* pp 4805–10
- [6] In H, Kang B B, Sin M and Cho K-J 2015 Exo-glove: a wearable robot for the hand with a soft tendon routing system *IEEE Robot Autom. Mag.* **22** 97–105
- [7] Polygerinos P, Wang Z, Overvelde J T B, Galloway K C, Wood R J, Bertoldi K and Walsh C J 2015 Modeling of soft fiber-reinforced bending actuators *IEEE Trans. Robot* **31** 778–89
- [8] Zhang J 2020 Modeling of a bending supercoiled polymer (SCP) artificial muscle *IEEE Robot Autom. Lett.* **5** 3822–9
- [9] Chou C-P and Hannaford B 1996 Measurement and modeling of McKibben pneumatic artificial muscles *IEEE Trans. Robot Autom.* **12** 90–102
- [10] Daerden F and Lefeber D 2002 Pneumatic artificial muscles: actuators for robotics and automation *Eur. J. Mech. Environ. Eng.* **47** 11–21
- [11] Al-Fahaam H, Nefti-Meziani S, Theodoridis T and Davis S 2018 Contractor pneumatic artificial muscle *Soft Robot* **5** 576–91
- [12] MacDonald B K, Cockerell O C, Sander J W A S and Shorvon S D 2000 The incidence and lifetime prevalence of neurological disorders in a prospective community-based study in the UK *Brain* **123** 665–76
- [13] Louis E D and Ferreira J J 2010 How common is the most common adult movement disorder? Update on the worldwide prevalence of essential tremor *Mov. Disorders* **25** 534–41
- [14] Thenganatt M A and Louis E D 2012 Distinguishing essential tremor from Parkinson's disease: bedside tests and laboratory evaluations *Expert Rev. Neurother.* **12** 687–96
- [15] Parkinson J 2002 An essay on the shaking palsy *J. Neuropsychiatr. Clin. Neurosci.* **14** 223–36
- [16] Daneault J-F, Carignan B and Duval C 2011 Finger tremor can be voluntarily reduced during a tracking task *Brain Res.* **1370** 164–74
- [17] Elble R J 2013 What is essential tremor? *Curr. Neurol. Neurosci. Rep.* **13** 353–60
- [18] Elble R J and Deuschl G 2009 An update on essential tremor *Curr. Neurol. Neurosci. Rep.* **9** 273–7
- [19] Louis E D 2005 Essential tremor *Lancet Neurol.* **4** 100–10
- [20] Rozman J, Bartolić A and Ribarić S 2007 A new method for selective measurement of joint movement in hand tremor in Parkinson's disease patients *J. Med. Eng. Technol.* **31** 305–11
- [21] Jankovic J, Schwartz K S and Ondo W 1999 Re-emergent tremor of Parkinson's disease *J. Neurol. Neurosurg. Psychiatry* **67** 646–50
- [22] Riviere C and Thakor N 1996 Modeling and canceling tremor in human-machine interfaces *IEEE Eng. Med. Biol. Mag.* **15** 29–36

- [23] Madere K 2016 Readi-steady orthotic glove system Readi-Steady (available at: <http://www.readi-steady.com/>) (Accessed March 2017)
- [24] GyroGear 2016 GyroGear Ltd (available at: <http://gyrogear.co/gyroglove>) (Accessed March 2017)
- [25] Kotovsky J and Rosen M J 1998 A wearable tremor-suppression orthosis *J. Rehabil. Res. Dev.* **35** 373–87
- [26] Trotman A 2017 How a watch helped Emma write again (available at: <https://news.microsoft.com/en-gb/features/how-a-watch-helped-emma-write-again/>) (Accessed May 2020)
- [27] Manto M, Rocon E, Pons J, Belda J M and Camut S 2007 Evaluation of a wearable orthosis and an associated algorithm for tremor suppression *Physiol. Meas.* **28** 415–25
- [28] Rocon E, Ruiz A F and Pons J L *et al* 2008 Wearable upper limb robots *Wearable Robots: Biomechatronic Exoskeletons* (Chichester, UK: Wiley) Pons J L pp 235–81
- [29] Sabari J, Stefanov D G, Chan J, Goed L and Starr J 2019 Adapted feeding utensils for people with Parkinson's-related or essential tremor *Am. J. Occup. Ther.* **73** 7302205120
- [30] Zhou Y, Jenkins M E, Naish M D and Trejos A L 2018 Development of a wearable tremor suppression glove *Proc. IEEE Int. Conf. Biomed. Robot Biomechatron. (BioRob) Enschede, Netherlands* pp 640–5
- [31] Yap H K, Kamaldin N, Lim J H, Nasrallah F A, Goh J C H and Yeow C-H 2016 A magnetic resonance compatible soft wearable robotic glove for hand rehabilitation and brain imaging *IEEE Trans. Neural Syst. Rehabil. Eng.* **25** 782–93
- [32] Yun -S-S, Kang B B and Cho K-J 2017 Exo-glove PM: an easily customizable modularized pneumatic assistive glove *IEEE Robot Autom. Lett.* **2** 1725–32
- [33] Sasaki D, Noritsugu T, Takaiwa M and Yamamoto H 2004 Wearable power assist device for hand grasping using pneumatic artificial rubber muscle *IEEE Int. Symp. Robot Hum. Interact. Commun. Okayama, Japan*
- [34] Hong T H and Park Y-L 2019 Design of pneumatic origami muscle actuators (POMAs) for a soft robotic hand orthosis for grasping assistance *Proc. IEEE Int. Conf. Soft Robot (RoboSoft) Seoul, Korea* pp 627–32
- [35] Cathers I, O'Dwyer N and Neilson P 2006 Entrainment to extinction of physiological tremor by spindle afferent input *Exp. Brain Res.* **171** 194–203
- [36] Kursda K, Lattanza L, Diao E and Rempel D 2006 In vivo flexor tendon forces increase with finger and wrist flexion during active finger flexion and extension *J. Orthop. Res.* **24** 763–9
- [37] Hussain Z, Azland N Z and Yusof A Z B 2018 Human hand motion analysis during different eating activities *Appl. Bionics Biomech.* **2018** 8567648
- [38] Wirekoh J and Park Y-L 2017 Design of flat pneumatic artificial muscles *Smart Mater. Struct.* **26** 035009
- [39] Wirekoh J, Valle L, Pol N and Park Y-L 2019 Sensorized, flat, pneumatic artificial muscle embedded with biomimetic microfluidic sensors for proprioceptive feedback *Soft Robot* **6** 768–77
- [40] Timoshenko S and Woinowsky-Krieger S 1959 Simply supported rectangular plates under the combined action of lateral loads and of forces in the middle plane of the plate *Theory of Plates and Shells* (New York: McGraw-Hill) pp 387–90
- [41] Timoshenko S and Woinowsky-Krieger S 1959 General equations for large deflections of plates *Theory of Plates and Shell* (New York : McGraw-Hill) pp 415–20
- [42] Jang S-H, Na S-H and Park Y-L 2017 Magnetically assisted bilayer composites for soft bending actuators *Materials* **10** 646
- [43] Mosadegh B, Polygerinos P, Keplinger C, Wennstedt S, Shepherd R F, Gupta U, Shim J, Bertoldi K, Walsh C J and Whitesides G M 2014 Pneumatic networks for soft robotics that actuate rapidly *Adv. Funct. Mater.* **24** 2163–70
- [44] Veale A J, Xie S Q and Anderson I A 2016 Characterizing the Peano fluidic muscle and the effects of its geometry properties on its behavior *Smart Mater. Struct.* **25** 065013
- [45] Abeach L A, Nefti-Maziani S, Theodoridis T and Davis S 2018 A variable stiffness soft gripper using granular jamming and biologically inspired pneumatic muscles *J. Bionic Eng.* **15** 236–46
- [46] Narang Y S, Aktas B, Ornellas S, Vlassak J J and Howe R D 2020 Lightweight highly tunable jamming-based composites *Soft Robot*
- [47] Rich S, Jang S-H, Park Y-L and Majidi C 2017 Liquid metal-conductive thermoplastic elastomer integration for low-voltage stiffness tuning *Adv. Mater. Technol.* **2** 1700179
- [48] Tonazzini A, Mintchev S, Schubert B, Mazzolai B, Shintake J and Floreano D 2016 Variable stiffness fiber with self-healing capability *Adv. Mater.* **28** 10142–8